MIL-I-83152B(USAF)
16 October 1974
SUPERSEDING
MIL-I-83152A(USAF)
5 September 1973

MILITARY SPECIFICATION

INDICATOR, INDICATED AIRSPEED AVU-22/A (INDICATOR-TRANSDUCER)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an integrally white lighted, 3-inch, round-dial, pitot-static, indicated airspeed indicator and transducer, designated AVU-22/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-B-5087	Bonding, Electrical, and Lighting Protection, for Aerospace Systems
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-C-83723	Connectors, Electric, Circular, Environment Resisting, General Specification for

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1524	Table of Differential Pressure in Relation to Calibrated Airspeed
MS28105	Cover Glass, Aircraft Instrument Dial
MS33556	Case, Aircraft Instrument, 3-1/4 Inch Size, Standard Dimensions for
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33649	Bosses, Fluid Connection - Internal Straight Thread

PUBLICATION

AFSC Design Handbook

DH 1-4	Electromagnetic Compatibility

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NASA Technical Note

D822 Tables of Airspeed, Attitude, and Mach Number Based on Latest International Values for Atmospheric Properties and Physical Constants

(Copies of the NASA document may be obtained from the National Aeronautics and Space Administration, Washington, DC 20546.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein. When materials are used that are not specifically designated, they shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Fungusproof materials. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

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3.3.4 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction

3.4.1 Design. The indicator shall be designed to provide an indication and an electrical output of indicated airspeed as a function of the difference between pitot and static pressures. The display design shall also include a maximum allowable airspeed pointer and preset Mach index. The maximum allowable airspeed pointer shall be mechanized to indicate either a preset maximum airspeed or the airspeed equivalent to the present Mach number, whichever is less. Adjustments shall be provided for setting both the maximum allowable airspeed pointer and the maximum allowable Mach number index. The indicator shall conform to figures 1 and 2. The electrical output characteristics shall be in accordance with table I. The design of the indicator shall be reasonably simple to permit overhaul or repair without requiring special tools or fixtures.

3.4.1.1 Reliability. The indicator shall have a minimum acceptable mean-time-between-failure (MTBF) of 1,200 hours.

3.4.1.2 Longevity. The indicator shall be designed for a useful life of 3,000 hours as defined by MIL-STD-781.

3.4.1.3 Electromagnetic interference. The generation of electromagnetic interference and susceptibility to interference from external sources shall be controlled in the equipment design. The design shall be governed by AFSC Design Handbook DH 1-4. The indicator shall be designed to meet the requirements of MIL-STD-461 when tested in accordance with the applicable test methods of MIL-STD-462.

3.4.1.3.1 Electrical bonding and grounding. The electrical bonding and grounding requirements for the indicator shall be in accordance with MIL-B-5087.

3.4.2 Construction. The indicator shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

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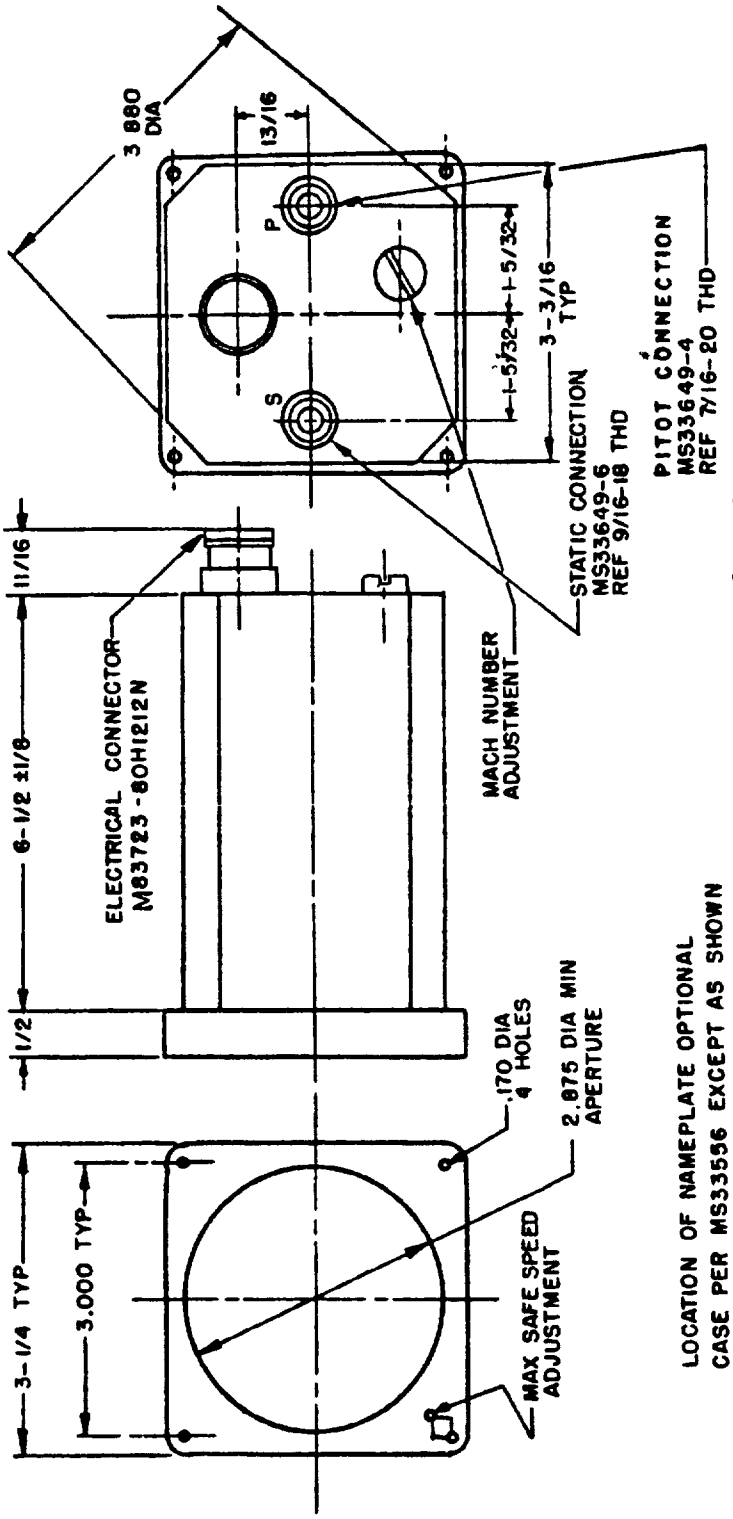
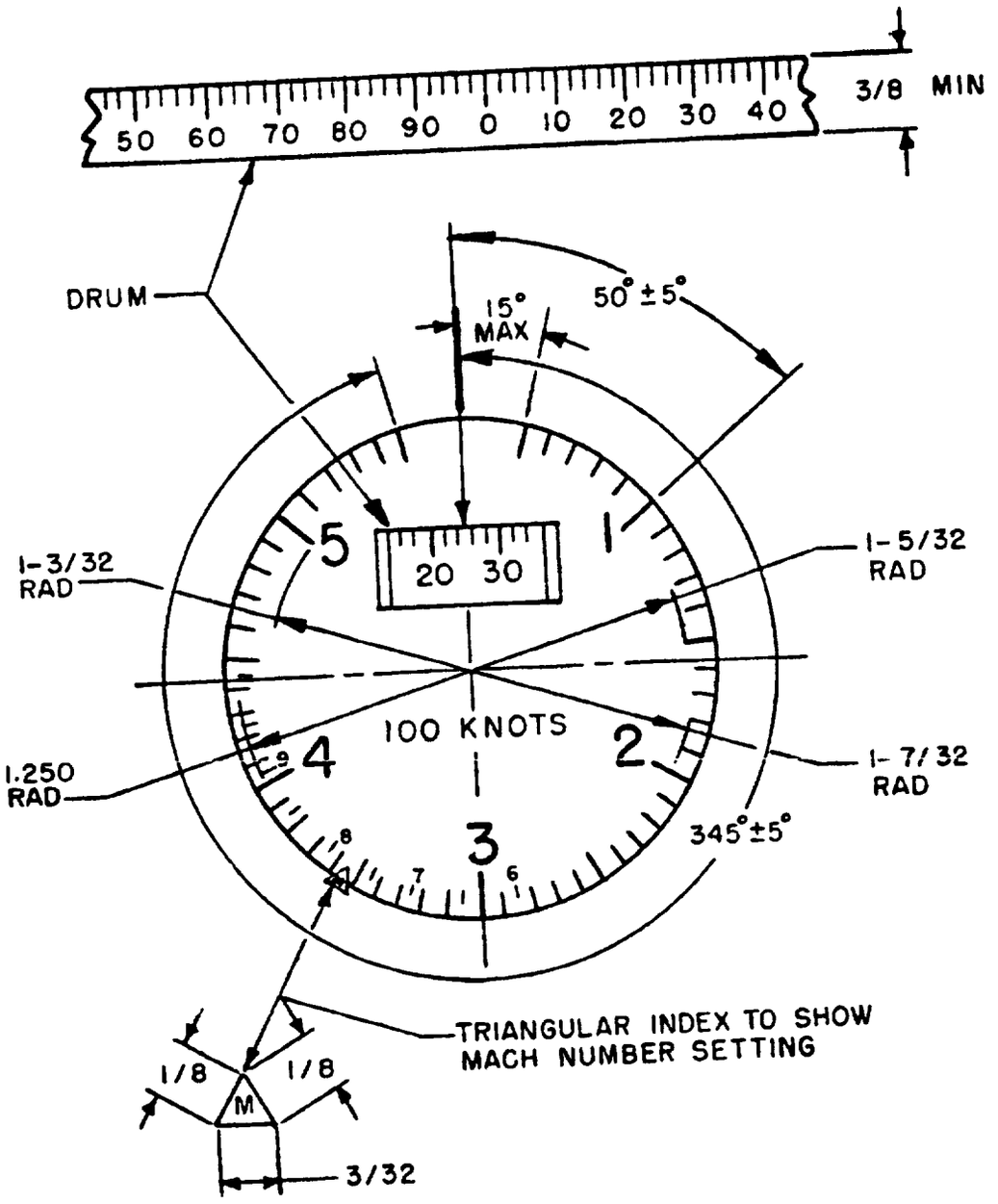


Figure 1. Case

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ALL GRADUATIONS ARE PARTIAL RADIAL LINES

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCES: FRACTIONS ±1/64 .

Figure 2. Dial

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TABLE I - Electrical Data

Synchro: Control transmitter

Scale rate: 334.285° for 500 knots V_c

0.66857°/knot

Electrical zero: As a CX at 50 knots ± 2 knots

Rotor excitation: 26V rms 400 Hz

Stator output: 3.9 max volt rms at 400 Hz

Line-to line open circuit at 57° phase

Detail data: @ + 25°C

a. Rotor resistance (DC) = 54 ohms ± 10%

b. Rotor impedance (Z) = 432 ohms /60° (216 + J302)

c. Stator resistance (DC) = 249 ohms ± 15%

d. Stator impedance (Z) = 208 ohms /30° (180 + J104)

e. Input current, rotor = 0.060 amps

f. Input power, rotor = 0.800 watts

Stator load: 10K min per leg matched

Display to electrical angle output accuracy: ±1.5 knots

Electrical zero procedure:

a. Tie C to Z

b. Apply 26V, 400 Hz to C and H

c. With a VTVM connected to X and Y, adjust the shaft position (relative to stator) to obtain a minimum voltage

d. Using a voltmeter, check voltage from X to H; if voltage is approx. 15V, turn shaft 180° relative to stator so that approx. 35V are measured from X to H

e. Re-establish null from X to Y

f. With a voltmeter across X to H, the voltage shall increase as the shaft is turned (less than 30°) CW for increasing airspeed. If the voltage decreases, the designations at X and Y shall be interchanged and the exact null re-established.

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3.5 Performance. The indicator shall be capable of meeting the requirements specified herein when subjected to the following conditions or any natural combination thereof:

a. Operating temperatures ranging from -54° to +71°C, and nonoperating temperatures ranging from -65° to +71°C

b. Vibration:

(1) Vibration failure: Vibration at a double amplitude of 0.1 each from 5 to 20 Hz; 2g from 20 to 500 Hz.

(2) Vibration error: Vibration with a double amplitude of between 0.003 and 0.005 inch at frequencies from 500 to 3,000 cpm.

c. Exposure to simulated salt-sea atmosphere for a period of 50 hours

d. Relative humidity up to 95 percent

e. Exposure to fungus growth as encountered in tropical climates

f. Applied acceleration forces up to 13.5g

g. Continuous use without change in calibration within the range of 40 to 550 knots

h. Overpressures as listed in table II

i. Exposure to sand and dust particles.

3.5.1 Magnetic effect. The indicator shall not cause a free compass to deflect more than 1° when revolved about a short-bar magnetic compass with a field intensity of 0.17 to 0.19 oersted.

3.5.2 Case leakage. Case leakage shall not exceed 0.1 inch Hg for a 1-minute period under the conditions specified in section 4.

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TABLE II. Test Tolerances

Test and Test Points	Tolerances
Position Error (50 to 550 knots)	
Airspeed pointer	3 knots
Maximum speed pointer	3 knots
Friction (50 to 550 knots)	
Maximum speed pointer	3 knots
Airspeed pointer	3 knots
Vibration Error	
(Mach number setting at 0.75)	
(Test altitudes of 0 and 20,000 feet)	
(Differential pressure equal to 100 and 400 knots)	
Maximum oscillation (both pointers)	2 knots
Variation from original position (both pointers)	2 knots 2 knots
Vibration Failure	
(Test altitude of 20,000 feet)	
(Differential pressure equal to 400 knots)	
(Mach number setting at 0.8)	2 knots
Overpressure	
(Static connection - 6 in. water)	
(Pitot connection - 30 in. Hg)	
(Test altitude of 0 and 20,000 feet)	
(Differential pressure equal to 100 and 100 knots)	
Low Temperature Operation (both pointers)	2 knots
High Temperature Operation (both pointers)	2 knots
Reliability (both pointers)	2 knots
Shelf Life	3 knots
Damping	
Time required for pointer to travel from full scale deflection to the 60 knots graduation	$1^{+1.0}_{-0.2}$ second

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3.5.3 Diaphragm capsule leakage. Diaphragm leakage shall not cause more than 1-knot deflection of the pointer for a 1-minute period under the conditions specified in section 4.

3.5.4 Scale, friction, and position errors. The scale, friction, and position errors shall not exceed the tolerances specified in tables II, III, and IV.

TABLE III. Airspeed Scale Error

Test Pressure Knots	Tolerance Knots
50*#	3.0
80	2.0
100*	2.0
120	2.0
150*#	2.5
180	2.5
200*	3.0
220	3.0
260*#	3.0
300	4.0
340*	4.0
380	4.0
400*#	5.0
420	5.0
460*	5.0
500	6.0
550*#	6.0

Note: * Represents friction test points.
Represents position error test points and low temperature test points (-54° ±2°C)

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TABLE IV. Mach Number

Altitude 1,000 Feet	0.6 Airspeed Knots	0.6 Tolerance ± Knots	0.7 Airspeed Knots	0.7 Tolerance ± Knots	0.8 Airspeed Knots	0.8 Tolerance ± Knots	0.9 Airspeed Knots	0.9 Tolerance ± Knots	1.0 Airspeed Knots	1.0 Tolerance ± Knots
0	397*#	4	463*	6	529*	8				
5			426	5	488	6				
10	333	4	390	4	448#	6	507	6		
15			356	4	410	5				
20	275*#	4	323*	4	375*	4	423*	5	475*	6
25			292	4	337	4				
30	223	4	262	4	303*	4	346	4	390	4
35			235	4	272	4				
40	177*#	4	209*	4	242*	4	277*	4	312*	4
45			186	4	216	4				
50	140	4	165	4	1920	4	220	4	248	4

Note: * Represents friction test points

Represents position error test points and low temperature test points (-54° ±2°C)

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3.6 Case. The indicator case shall conform to MS33556 for the 3-inch nominal size, except as shown on figure 1.

3.6.1 Body. The body shall be made of aluminum alloy, shall be uniform in texture with a smooth surface, and shall be entirely covered with a lusterless black finish conforming to color No. 37038 of FED-STD-595.

3.6.2 Bezel ring. The bezel ring shall be made of aluminum alloy and shall have a durable, lusterless-black finish. The ring shall be held in place by means of screws properly secured by lockwashers.

3.6.3 Cover glass. The cover glass shall conform to MS28105-7 and shall be properly sealed into the bezel ring. The bezel shall be held in place by means of screws properly secured by lockwashers or similar devices of approved design. If the cover glass is used as part of the lighting system, a glass may be used with characteristics deviating from the glass specified herein to the extent required for the lighting function provided it does not interfere with the readability of the instrument.

3.6.3.1 Coating of cover glass and lighting wedge. Coating of the cover glass, lighting wedge, and any other reflecting glass surfaces shall be in accordance with MIL-C-14806.

3.6.4 Dial. The dial shall be fastened securely into the case or upon the frame of the mechanism by at least two screws in such a manner that it will not loosen or turn when the indicator is vibrated. The dial and drum shall conform to figure 2. The drum shall make 1 revolution for each 100-knot change in airspeed and shall be so geared to the indicated airspeed pointer that proper relationship is maintained at all times.

3.6.4.1 Dial and cover glass location. The distance between the front surface of the bezel and the outside surface of the cover glass shall not exceed 0.062 inch. The distance between the inside surface of the cover glass and the dial shall not exceed 0.125 inch.

3.6.5 Index. A triangular index as shown on figure 2 shall be provided to show the maximum allowable Mach number to which the instrument is adjusted. The Mach numbers shall be placed on the dial as shown on figure 2, and the mechanism shall be so arranged that the values of the Mach number between the numerals may be set by linear interpolation between the numerals.

3.6.6 Maximum allowable Mach number adjustment. The maximum allowable Mach number adjustment shall be accessible by means of a plug located at the rear of the indicator case. The range of adjustment shall be between 0.6 and 1.0 Mach number. Adjustment shall be made with the use of standard tools.

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3.6.7 Constant maximum allowable airspeed adjustment. The constant maximum allowable airspeed shall be adjustable from the front of the indicator by a screwdriver slot in the lower left corner of the case. The range of adjustment shall be from 300 to 550 knots.

3.7 Pointers. The pointers shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. They shall be firmly attached to the mechanism and shall be readily adjustable. The indicated airspeed pointer shall conform to MS33585-3 and shall be finished in lusterless white, color No. 37875 of FED-STD-595. An MS33585-3 maximum airspeed pointer shall be provided, except that the shaded portion of the pointer shall be finished as shown on figure 3, and the pointer width shall be 0.150 inch in lieu of 0.093 inch. The nonindicating ends of the pointers shall not obscure the drum. The two pointers shall be concentric, with the maximum speed pointer closest to the dial and the indicated airspeed pointer nearest the cover glass.

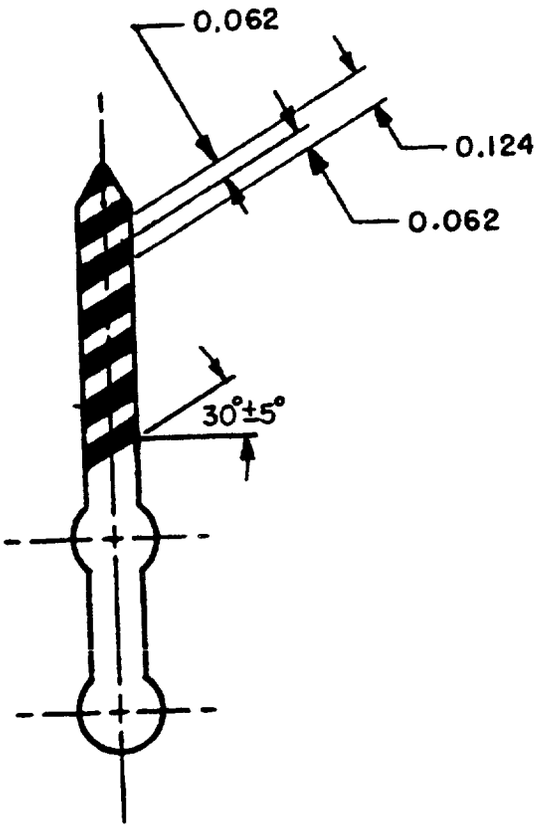
3.7.1 Indicated airspeed stops. The indicated airspeed pointer movement shall be limited by suitable stops in the mechanism in such a way that the pointer will not be permitted to rotate more than 358° of arc clockwise from the no-pressure position when the indicator is subjected to the overpressure tests.

3.8 Electrical connector. The electrical connector shall be an M83723-80H 1212N connector with a hermetic seal, solder flange mount, and a bayonet coupling in accordance with MIL-C-83723.

3.9 Integral lighting. The indicator shall be integrally lighted with white lighting circuits as shown on figure 4. The white lighting shall be in accordance with MIL-L-27160. Lighting shall operate on 5V ac or dc. Lighting acceptability of production indicators shall be based on visual comparison with a prime standard. The method of lighting shall be approved by the procuring activity prior to fabrication.

3.9.1 Prime standard. The prime standard lighting reference instrument shall be prepared as follows: A minimum of 20 areas covering the entire face of the indicator shall be measured in foot-lamberts for brightness level. A minimum of 10 areas covering the entire face of the indicator shall be measured for color. The prime standard instrument and its recorded overall brightness and color measurements shall be approved by the procuring activity. The standard shall be checked at least every 50 hours of operation or 120 days, whichever occurs first, to determine any change in brightness or color. Variations greater than ±20 percent in two or more areas shall require return of the standard for corrective action.

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SHADED AREA TO BE LUSTERLESS WHITE, COLOR NO.37875 OF FED-STD-595.

UNSHADED AREA TO BE FINISHED IN DURABLE LUSTERLESS BLACK, COLOR NO.37038 OF FED-STD-595.

Figure 3. Maximum Airspeed Pointer

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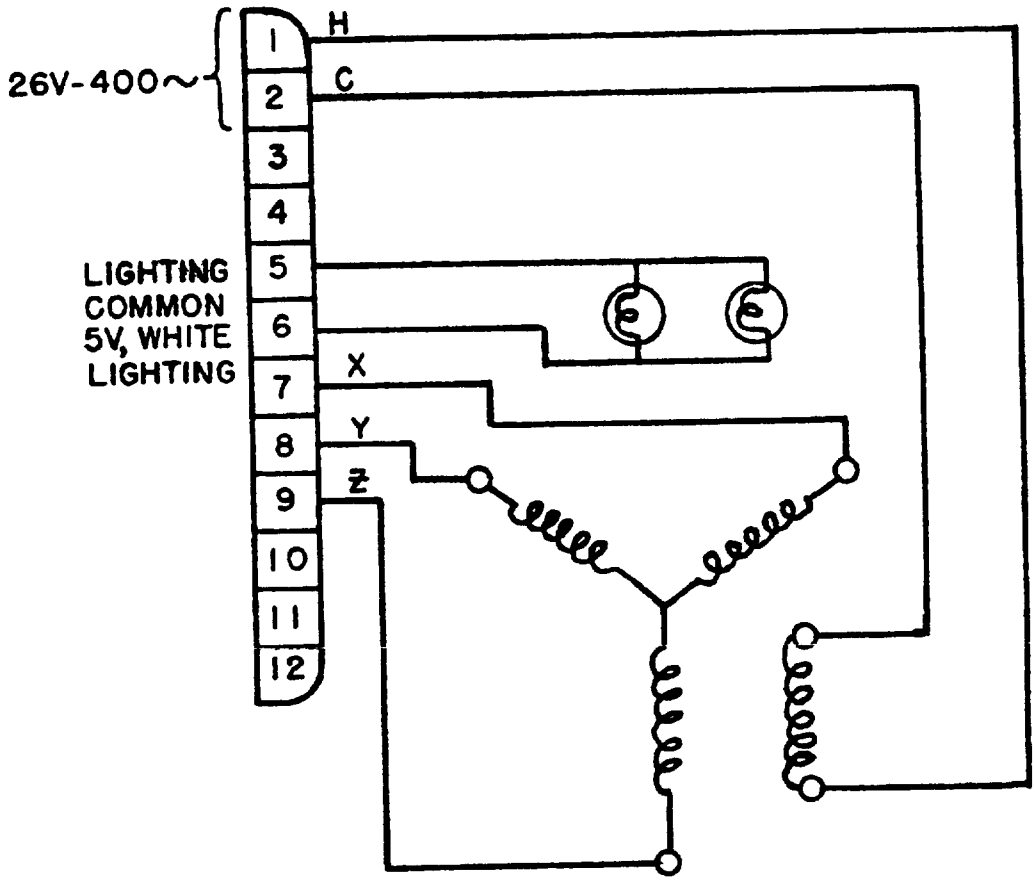


Figure 4. Electrical Circuits

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3.10 Application of pressure. The pitot pressure shall be applied to the inside of the diaphragm capsule, and the static pressure shall be applied to the interior of the case.

3.10.1 Capillary tubing. A sufficient length of approved capillary tubing having a minimum inside diameter of 0.020 inch, or some other acceptable restriction, shall be incorporated in the instrument case between the pitot connection in the rear of the case and the diaphragm capsule. The effect of this restriction shall be such that the indicator will meet the damping test specified herein.

3.10.2 Pressure fittings. Pitot and static pressure fittings shall be as shown on figure 1.

3.10.3 Plug. During shipment, each static and pitot pressure connection shall be provided with a vented plug of a material and design acceptable to the procuring activity. The vent hole shall be 1/16±1/64 inch is diameter.

3.10.4 Calibration and calibration standards. The indicator shall be calibrated throughout its operating range in accordance with MIL-STD-1524. Defining equations shall be in accordance with NASA Technical Note D822.

3.11 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.12 Markings. All markings shall be durable to withstand the usage encountered in service. The style and proportions of numerals and letters placed on the dial shall conform to MS33558. Numerals shall distinctly indicate the graduations to which each applies. The dimensions of the dial markings shown in table V shall be lusterless white, color No. 37875 of FED-STD-595.

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TABLE V. Dial Markings

	Height or Length	Width of Line or Grad
Graduations corresponding to numerals 50, 150, 250 350, 450, 550 knots	As shown on figure 2	0.031
Numerals 1, 2, 3, 4, 5	5/16	0.031
10 knot graduations	See figure 2	0.016
Triangular index	See figure 2	
Graduations on drum corresponding to numerals	1/8	0.031
2 knot graduations on drum	3/32	0.016
Numerals on drum	3/16	0.016
Lettering 100 KNOTS		

3.12.1 Dial. The markings USAF "AVU-22/A" and AIRSPEED shall be permanently marked on the dial in letters 3/64 ±1/64 inch high. These markings and all other markings not otherwise specified shall be durable lusterless black, color No. 37038 of FED-STD-595.

3.12.2 Mach index. The following markings shall be finished in gloss black, color 17038 of FED-STD-595. The marking dimensions shall be as follows:

	Height of Length Inch ± 0.010	Width of Line or Graduation Inch ± 0.005
Triangular index for Mach No. setting	As shown on figure 2	As shown on figure 2
Graduations corresponding to Mach numerals	0.062	0.016
Numerals representing Mach No. settings	0.093	0.016

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3.12.3 Dial marking visibility. At least 1/16 inch of the graduations, numerals, pointers, and other markings on the dial shall be visible from any point within the frustum of a cone the side of which makes an angle of not less than 20° with the perpendicular to the dial and of which the small diameter is the aperture of the dial.

3.12.4 Back markings. The back of the case, adjacent to the connections, shall be marked as follows:

P	Pitot connection
S	Static connection
MACH NUMBER ADJUSTMENT	Mach number adjustment cover

3.13 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.14 Weight. The weight of the indicator shall not exceed 3.00 pounds.

3.15 Finishes and protective coatings

3.15.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.15.1.1 Dials, small holes, and case inserts need not be anodized.

3.15.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.15.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.15.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.15.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.15.2 Steel parts. Where practicable, steel parts shall be cadmium plated in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

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3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.17.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.17.2 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.17.3 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered, or excess solder; metal chips, and other foreign material after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Quality conformance tests . . . . . See 4.5.

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4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of test are not specified definitely, it shall be understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Tapping. Unless otherwise specified, the indicator shall be vibrated before a test reading is taken. Vibration shall be applied by electromechanical vibrators and shall be in the order of 0.2g. Frequencies of less than 3,000 cpm shall not be used.

4.3.3 Attitude. Unless otherwise specified, the indicator shall be tested in its normal operating position.

4.3.4 Vibration. A vibration stand shall be used that will vibrate at any desired frequency between 500 and 3,000 cpm and will subject the indicator to such vibration that a point on the case will describe, in a plane inclined 45° to the horizontal plane, a circle of the size diameter specified herein.

4.3.5 Pointer stop adjustment. The maximum speed pointer stop shall be so adjusted that the maximum speed pointer will not stop until 550 knots is reached.

4.3.6 Airspeed drum. When tests on the airspeed pointer are specified, the tests shall include the airspeed drum.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of nine indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.2.1).

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6. The test samples specified in 4.4.1 shall be subjected to the preproduction tests as follows:

Group I (3 Indicators)	Group II (6 Indicators)
All tests, except Reliability and Shelf-Life	Reliability and Shelf-Life only

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4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described in 4.6:

a. Examination of product

b. Case leakage

c. Diaphragm capsule leakage

d. Airspeed scale error at room temperature

e. Airspeed transducer output scale error at room temperature

f. Maximum speed scale error at room temperature

g. Hysteresis

h. After effect

i. Friction airspeed pointer

j. Friction maximum speed pointer

k. Individual tests of MIL-L-27160

l. High temperature operation.

4.5.2 Sampling tests

4.5.2.1 Sampling plan A. One indicator shall be selected at random from each one hundred or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Case leakage at -35°C

c. Position error

d. Low temperature operation

e. Vibration error

f. Sampling plan A tests of MIL-L-27160

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4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), three indicators shall be selected at random from the first fifteen on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Low temperature exposure

c. High temperature exposure

d. Acceleration

e. Humidity

f. Salt-fog

g. Vibration failure

h. Fungus

i. Damping

j. Magnetic effect

k. Pointer zero position error

l. Overpressure

m. Dust

n. Rain

o. Sampling plan B tests of MIL-L-27160

p. Reliability and shelf life

4.5.2.2.1 Rejection and retest. When one indicator selected from a production run fails to meet the specification, items still on hand or later produced shall not be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.2.1.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the indicators on hand or items produced later shall not be made until it is determined that all items meet all the requirements of the specification.

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4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in indicators already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Case leakage. The pitot and static pressure connections of the indicator shall be joined with a Y connection and connected to the same mercury manometer and a source of vacuum and pressure. A suction of 25 inches Hg shall be applied to the pitot and static connections of the indicator. With the source disconnected for a period of 1 minute, the difference in the mercury levels shall not change by more than 0.1 inch.

4.6.2.1 Case leakage at -35°C. The indicator shall be placed in a chamber and maintained at a temperature of -35°C for a period of 3 hours. While at this temperature, a case leakage test in accordance with 4.6.2 shall be conducted.

4.6.3 Diaphragm capsule leakage. With the static connection open to the atmosphere, a pressure sufficient to produce approximately full-scale deflection of the pointer shall be applied to the pitot connection of the indicator at which point the connecting tubing shall be pinched off or otherwise completely sealed. During a period of 1 minute, the pointer shall not change its position by more than 1 knot.

4.6.4 Airspeed scale error at room temperature. The indicator shall be tested for scale error (see 6.3.1) at the points on the scale shown in table III. The test shall be made by subjecting the indicator to the pressures specified to produce these readings, first with increasing pressures, then with decreasing pressures. For increasing pressures, the pressure shall be brought up to but shall not exceed the pressures specified to give the desired reading and, for decreasing pressure, the pressure shall be brought down to but not below the pressures specified to give the desired reading. The errors at the test points shall not exceed those specified in table III.

4.6.4.1 Airspeed transducer output scale error at room temperature. In conjunction with the indicator scale error test, the transducer output shall also be measured. At each of the test points specified in table III, the transducer output shall agree within 1.5 knots of the airspeed pounter indication.

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4.6.4.2 Maximum speed scale error at room temperature. The maximum speed pointer shall be tested for scale errors at the points of the scale indicated by single asterisks in table IV. The test shall be accomplished by adjusting the Mach number adjustment to the setting listed at the top of table IV. During this test, the pitot and static connection shall be connected together to a standard mercury barometer and to sources of vacuum and pressure. The reduction in pressure, as indicated by the barometer, shall be made at a rate corresponding to an increase in altitude of approximately 6,000 fpm. The scale errors shall not exceed the tolerances specified in table IV. This test shall be combined with the hysteresis test at each Mach number setting.

4.6.4.3 Hysteresis. Not more than 15 minutes after the indicator has been subjected to the pressure corresponding to the highest altitude specified in table IV, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 3,000 fpm until the pressure corresponding to 30,000 feet is reached. The indicator shall remain at this pressure for at least 5 minutes, but not more than 15 minutes, before the test reading is taken. After the reading has been taken, the pressure shall be further increased at the above rate until the pressure corresponding to 10,000 feet is reached. The indicator shall remain at this pressure for at least 1 minute, but not more than 10 minutes, before a test reading is taken. After the reading has been taken, the pressure shall be further increased at the above rate until atmospheric pressure is reached. The reading of the maximum speed pointer at either of the two test points shall not differ from the corresponding reading, with decreasing pressure, by more than 2 knots.

4.6.4.4 After effect. Not less than 1 minute, and not more than 5 minutes, after the completion of the maximum speed scale at room temperature and hysteresis tests, the maximum speed pointer shall be checked to determine that it has returned to its original reading, corrected for any change in atmospheric pressure, within 4 knots.

4.6.5 Friction airspeed pointer. The indicator shall be tested for friction at the single-asterisk test points of table III. The pressure shall be increased to bring the pointer to the approximate desired reading and held constant while two readings are taken, the first before the indicator is tapped, the second after the indicator is tapped. The difference between any two such readings shall not exceed the values specified in table II. The pointer shall move smoothly while the pressure is varied uniformly without vibrating the indicator. This test may be combined with the scale error at room temperature test.

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4.6.5.1 Friction maximum speed pointer. The indicator shall be tested for friction at each of the points marked with an asterisk in table IV. The pressure shall be decreased to bring the pointer to the desired reading and held constant while the two readings are taken, the first before the indicator is tapped and the second after the indicator is tapped. The difference between any two readings shall not exceed the tolerance specified in table II. The pointer shall move smoothly while the pressure is varied uniformly without vibration of the indicator. This test may be combined with the scale error test.

4.6.6 Position error indicated airspeed pointer and maximum speed pointer. The pointer reading, taken while the indicator is held in any desired position, and while it is being tapped, shall not differ from its reading, when held in normal operating position, by more than the amount specified in table II. At least one of these positions shall be at approximately 45° with the horizontal, simulating a 45° climb or dive or both. This test shall be conducted at the test points marked with a double asterisk in table III and IV. This test may be combined with the friction scale error test.

4.6.7 Low temperature operation. The indicator shall be placed in a chamber and maintained at a temperature of -54° ±2°C for a period of 4 hours. At the end of the 4-hour period and while at the low temperature, the indicator shall be subjected to scale error tests in accordance with 4.6.4, 4.6.4.1, and 4.6.4.2. The errors at the test points specified in tables III and IV shall not exceed those specified in tables III and IV by more than the values specified in table II. The airspeed transducer output shall agree within 2.0 knots of the airspeed pointer indication. No damage to any part of the indicator shall result from this test.

4.6.8 High temperature operation. The indicator shall be placed in a chamber and maintained at a temperature of 71° ±2°C for a period of 4 hours. At the end of the 4-hour period and while still at the high temperature, the indicator shall be subjected to scale error tests in accordance with 4.6.4, 4.6.4.1 and 4.6.4.2. The errors at the test points shall not exceed those specified in tables III and IV by more than the values specified in table II. The airspeed transducer output shall agree within 2.0 knots of the airspeed pointer indication. No damage to any part of the indicator shall result from this test.

4.6.9 Vibration error. The indicator shall be mounted on the vibration stand in its normal operating position and secured by flange screws positioned as on an aircraft panel. The indicator shall be subjected to vibration with an amplitude between 0.003 and 0.005 inch at frequencies from 500 to 3,000 cpm, so that any point on the indicator case will describe the circular path of amplitude specified, and inclined at an angle of 45° to the horizontal and dial planes. The natural frequency of the airspeed indicator shall not occur in this range While the indicator is being vibrated at test points specified in table II, the maximum total spread of pointer vibration and the pointer variation from its original position shall not exceed the values specified in table II.

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4.6.10 Low temperature exposure. The indicator shall be subjected to a low temperature test in accordance with MIL-STD-810, Method 502 for a period of 48 hours at -65° ±2°C. At the end of the 48-hour period, the temperature shall be raised to -54° ±2°C and shall be maintained at this temperature for 4 hours. While still at this temperature, the indicator shall be tested for scale errors at the test points specified in tables III and IV. The scale errors at the tests points shall not exceed those specified in tables III and IV by more than the values specified in table II. The airspeed transducer output shall agree with the airspeed pointer indication within 2.0 knots. After the indicator has returned to room temperature, it shall meet the individual tests except for hysteresis, after effect, and position error.

4.6.11 High temperature exposure. The indicator shall be subjected to a high temperature test in accordance with MIL-STD-810, Method 501, procedure I except for a period of 24 hours. After the indicator has returned to room temperature, it shall meet the individual tests, except for hysteresis and after effect.

4.6.12 Acceleration. The acceleration test shall be performed in accordance with MIL-STD-810, Method 513.1, procedures I and II. The indicator shall be mounted in its normal operating position with the rear of the case toward the center of rotation and shall be subjected to the acceleration specified in 3.5(f) for the appropriate axes in accordance with Method 513.1 procedure I for a period of 1 minute. The tests shall be repeated with the instrument rotated 90° in azimuth with respect to the normal position and with the dial horizontal. The airspeed pointer shall not vary by more than 2.0 knots during acceleration. After testing, the indicator shall be subjected to and shall meet the individual tests, except for hysteresis and after effect.

4.6.13 Humidity. The indicator shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507, procedure I. The indicator shall then be subjected to and shall meet the individual tests, except for hysteresis and after effect.

4.6.14 Salt-fog. The indicator shall be subjected to a salt-fog test in accordance with MIL-STD-810, Method 509, procedure I, for a period of 50 hours. At the end of the 50-hour period, there shall be no damage to any part of the indicator that would affect the performance of the instrument. The indicator shall then be subjected to and shall meet the individual tests, except for hysteresis and after effect.

4.6.15 Vibration failure. The vibration failure tests shall be conducted in accordance with MIL-STD-810, Method 514.1, equipment category (a), procedure I, part 1, figure 514.1-1, curve B at frequencies from 5 to 500 Hz.

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4.6.16 Fungus. The indicator, including applicable external connectors, shall be subjected to a fungus test in accordance with MIL-STD-810, Method 508. There shall be no deterioration that would adversely affect subsequent operation. The indicator shall then be subjected to and shall meet the individual tests, except for hysteresis and after effect.

4.6.17 Damping. Sufficient pressure shall be applied to the pitot connection to produce full-scale deflection of the airspeed pointer. The pressure shall suddenly be released. The time required for the airspeed pointer to reach the value specified in table II shall be equal to the time and within the tolerance specified therein.

4.6.18 Magnetic effect. The indicator shall be held in various positions with its nearest part 5 inches from the center of a short-bar magnetic compass with compensator magnets removed and which is set up in a uniform magnetic field having a horizontal intensity between 0.17 and 0.19 oersted. The various positions shall be so chosen as to include that position which causes the maximum deflection of the compass. The maximum deflection shall not exceed 1°.

4.6.19 Pointer zero position error. The zero position of the pointer with no differential pressure applied to the indicator shall be within the values specified in table II.

4.6.20 Overpressure. The pressure specified in table II shall be applied separately to the pitot pressure and static connections of the indicator for a period of 10 minutes each. Not less than 5 minutes following the applications of overpressure, the indicator shall be tested for scale error as specified in the scale error at room temperature test. The errors at the test points shall not exceed those specified in tables III and IV by more than the values specified in table II.

4.6.21 Dust. The indicator shall be subjected to a dust test in accordance with MIL-STD-810, Method 510, procedure I. The indicator shall then be subjected to and shall meet the individual tests, except for hysteresis and after effect.

4.6.22 Rain. The indicator shall be subjected to a rain test in accordance with MIL-STD-810, Method 506, procedure I. The indicator shall then be subjected to and shall meet the individual tests, except for hysteresis and after effect.

4.6.23 Lighting. Indicator lighting shall be tested in accordance with MIL-L-27160.

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4.6.24 Reliability. The six samples specified in 4.4.3 shall be subjected to a reliability test in accordance with MIL-STD-781, test level F, test plan IV, except as modified herein.

a. Each indicator shall be subjected to a pressure cycling of a minimum of 3 cycles per hour consisting of a pressure variation equivalent to approximately 3/4 of full scale. Each pressure cycle shall consist of operation to maximum required pressure and return to ambient. The maximum scale value shall be maintained for not less than 2 minutes. Approximately 2 minutes of ambient pressure shall be allowed between each pressure cycle.

b. After each 100 hours of testing, the indicator shall be subjected to and shall meet the tests specified in 4.6.1, 4.6.4, 4.6.4.1, 4.6.4.2, and 4.6 5.

c. Upon completion of the final cycle, the indicator shall be rested at ambient pressure for a period of 48 hours. The indicator shall then be subjected to and shall meet the individual tests. Failure to meet any test shall be considered a malfunction. In case of malfunction of any kind, the malfunction shall be recorded and the reliability test interrupted while repairs are made. The test shall then be continued from the point of interruption. If recalibration is required to pass any scale error test, an additional malfunction shall be charged against this test.

d. A failure shall be charged for any out-of-tolerance condition of less than 2.0 knots only if this condition exists at the same point in the subsequent test.

4.6.24.1 Shelf life. After completion of the reliability test, the indicators shall be placed in storage for a period of not less than 30 days. During the 30-day period, the indicators shall remain as dormant as practicable. The mechanism shall not be exercised other than that caused by normal ambient pressure vibrations. The temperature shall remain essentially constant; however, normal room temperature variations will be permissible. After the 30-day period, the indicators shall again be subjected to and shall meet the individual tests. In addition to meeting the tolerances specified for scale error, the readings shall not vary from the original scale error readings by more than the values specified in table II.

4.6.25 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471 and shall be performed on a minimum of three selected repairs. The tasks shall be selected in advance of timing the tasks and shall be approved by the procuring activity.

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4.6.26 Electromagnetic interference. Compliance with the applicable requirements of MIL-STD-461 shall be demonstrated by testing of the indicator in accordance with the following MIL-STD-462 test methods:

a. CE03 Conducted Emissions test

b. CS01 Conducted Susceptibility test

c. CS02 Conducted Susceptibility test

d. CS06 Conducted Susceptibility test

e. RE02 Radiated Emissions test

f. RS03 Radiated Susceptibility test

4.6.26.1 Bonding. Resistance tests shall be conducted between each indicator and its simulated aircraft mount to determine compliance with 3.4.1.3.1.

4.7 Inspection and test. Test for methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to assure compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

* 5.1.1 Level A. Indicators shall be preserved and packaged in accordance with MIL-STD-794 and Method III of MIL-P-116, one each in a unit container conforming to PPP-B-636, weather-resistant class. Electrical receptacles and connectors shall be equipped with water-vaporproof caps.

* 5.1.2 Level C. Indicators shall be individually preserved and packaged in a manner that will provide protection against corrosion, deterioration, or physical damage so that serviceability is assured during shipment from supply source to the first receiving activity. This level may be the contractor's commercial practice provided the latter meets the criteria for this level as defined in MIL-STD-794.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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* 5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed for consolidation in shipping boxes conforming to PPP-B-636, weather-resistant class. Exterior shipping containers shall be of uniform shape and size, of minimum tare, weight, and cube.

* 5.2.2 Level B. Indicators, preserved and packaged as specified in 5.1.1 shall be packed for consolidation in exterior shipping boxes conforming to PPP-B-636, domestic class. Other requirements as specified in 5.2.1 shall apply.

* 5.2.3 Level C. Indicators shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery, and shall meet as a minimum, the requirements of carrier rules and regulations. This level may be the contractor's commercial practice provided the latter meets the criteria for this level as defined in MIL-STD-794.

* 5.3 Marking. In addition to the standard marking requirements of MIL-STD-129, level C/C packs containing item(s)/equipment intended for immediate use at destination shall be conspicuously marked as follows on two opposite sides of the exterior container in letters as large as space permits:

FOR IMMEDIATE USE
DO NOT STORE IN THIS PACK

5.3.1 Nomenclature. Nomenclature shall be as follows:

INDICATOR, INDICATED AIRSPEED AVU-22/A
(INDICATOR-TRANSDUCER)

6. NOTES

6.1 Intended use. The AVU-22/A indicator is intended for use in aircraft to indicate the speed relative to the air at sea level and the equivalent sea level speed corresponding to the actual sustaining force of the air at any other altitude. It will also indicate the maximum allowable airspeed and provide an electrical output.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B tests will not be conducted (see 4.5.2.2)

c. Levels of preservation, packaging and packing required (see 5.1 and 5.2)

d. Special shipment marking (see 5.4).

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6.2.1 Test report. The contractor will submit three copies of the test report specified in 4.4.2 to the procuring activity for approval.

6.3 Definition

6.3.1 Scale error. Scale error is the difference between the indication of the instrument and the calibrated airspeed corresponding to the applied differential pressure.

6.4 External connections. When external connections to simulate installed conditions are called for, they will consist of 10 feet of 1/4-inch copper or aluminum tubing so coiled that moisture can drain out the open end. Similar coils should be connected to the pitot and static pressure connections.

6.5 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Review activity:
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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

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